

Groups and Coding

Lecture—7-8

# ALGEBRAIC STRUCTURE



# Message and Word

Message: the basic unit of information, called message,

It is defined as a sequence of characters from a finite alphabet, where finite alphabet is  $B\{0,1\}$ .

Word: A word is a sequence of  $m$  elements from  $B$  ( 0's and 1's) also word is the basic unit of information.

The set B is a group under the binary operation +

| + | 1                                                                                               | 0 |
|---|-------------------------------------------------------------------------------------------------|---|
| 0 | 1                                                                                               | 0 |
| 1 | <del>0</del>  | 1 |

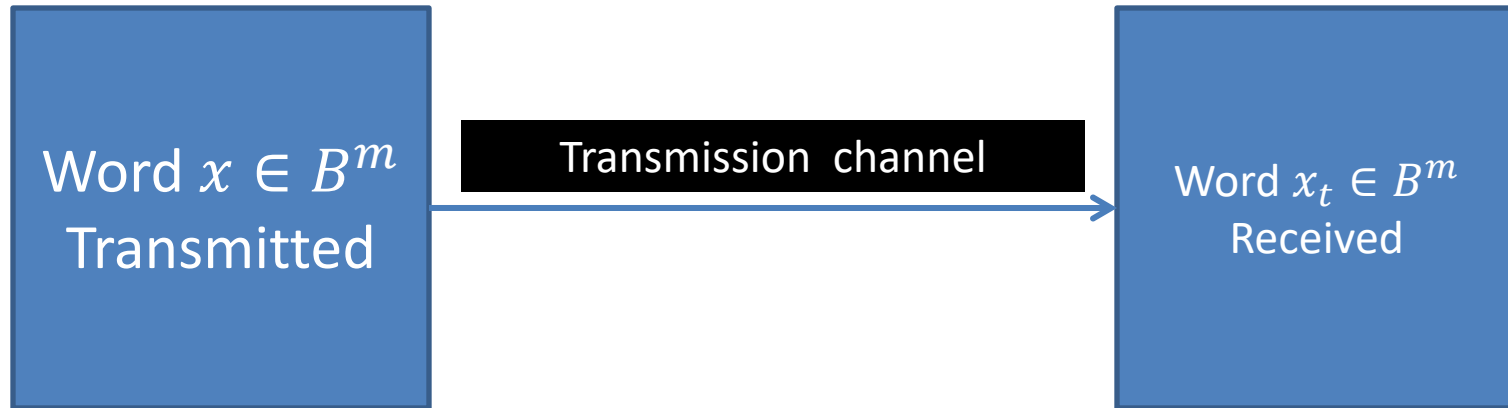
The set B  $\{0,1\}$  is a group under addition modulo 2

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |
|  |  |  |

Note 1: if  $B$  is a group then  $B^m = B \times B \times \cdots \times B$  ( $m$  factor) is a group under the operation  $\oplus$  defined by  $(x_1, x_2, \dots, x_m) \oplus (y_1, y_2, \dots, y_m) = (x_1 + y_1, x_2 + y_2, \dots, x_m + y_m)$

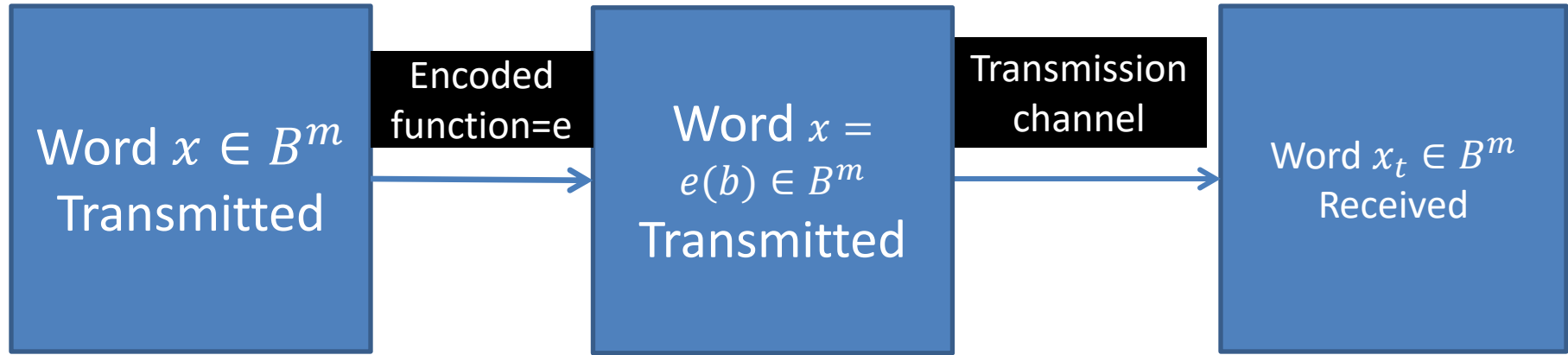
Note 2: The order of group  $B^m = 2^m$

# Noise:



Noise transmission if  $x \neq x_t$  this is due to weather interference , electrical problems and so on

## Reduction of Noise:



The function  $e$  is called an  $(m, n)$  encoding function

Weight: for  $x \in B^n$ , the number of 1's in  $x$  is called the weight of  $x$  and is denoted by  $|x|$

- Example: find the weight of each of the following words in  $B^5$ :
- (a)  $x=01000$
- (b)  $x=11100$
- (c)  $x=00000$

Ans: (a) 1, (b) 3 , (c)=0

Parity check code: the following encoding function  $e: B^m \rightarrow B^{m+1}$  is called the parity check code if  $b = b_1, b_2, b_3 \dots b_m \in B^m$  define

$$e(b) = b_1, b_2, \dots, b_m, b_{m+1}$$

$$\text{where } b_{m+1} = \begin{cases} 0 & \text{if } |b| \text{ is even} \\ 1 & \text{if } |b| \text{ is odd} \end{cases}$$

For example: for  $m=3$  then

$$e(000)=0000$$

$$e(001)=0011$$

$$e(010)=0101$$

$$e(011)=0110$$

$$e(100)=1001$$

$$e(101)=1010$$

$$e(110)=1100$$

$$e(111)=1111$$

00

011

01

0110

1001

1010

$$\left\{ \begin{array}{l} e(110) = 1100 \\ e(111) = 1111 \end{array} \right.$$

let encoding function  $e: B^m \rightarrow B^{3m}$  is defined  
as follow if  $b = b_1, b_2, b_3 \dots b_m \in B^m$  define

$$e(b) =$$

$$\underline{b_1, b_2, b_3 \dots b_m} \underline{b_1, b_2, b_3 \dots b_m} \underline{b_1, b_2, b_3 \dots b_m}$$

then for  $m=3$  find  $e(b)$

For example: for  $m=3$  then

$$e(000)=0000$$

$$e(001)=0011$$

$$e(010)=0101$$

$$e(011)=0110$$

$$e(100)=1001$$

$$e(101)=1010$$

$$e(110)=1100$$

$$e(111)=1111$$

$$e(000) = 0000000000$$

$$e(001) = 001001001$$

$$e(010) = 010010010$$

$$e(011) = 011011011$$

$$= 011011011$$

For example: for  $m=3$  then

$$e(000)=0000$$

$$e(001)=0011$$

$$e(010)=0101$$

$$e(011)=0110$$

$$e(100)=1001$$

$$e(101)=1010$$

$$e(110)=1100$$

$$e(111)=1111$$

Hamming Distance:  $\delta\{x, y\}$ : let  $x, y$  be words in  $B^m$ . The following distance  $\delta\{x, y\}$  between  $x, y$  is the weight,  $|x \oplus y|$  of  $x \oplus y$ . Thus the distance between  $x = x_1x_2x_3 \dots x_m$  and  $y = y_1y_2 \dots y_m$  is the number of values of  $i$  s.t  $x_i \neq y_i$  that is, the number of positions in which  $x$  and  $y$  differ using the weight of  $x \oplus y$  is a convenient way to count the number of different positions.

Example: Find the Hamming distance between  $x$  and  $y$ ;

$$(a) x = 110110 \quad ; y = 000101$$

$$(b) x = 001100 \quad ; y = 010110$$

Example: Find the Hamming distance between  $x$  and  $y$ ;

(a)  $x = 110110$  ;  $y = 000101$

(b)  $x = 001100$  ;  $y = 010110$

$$\delta(x, y) = |x \oplus y|$$

(a)  $x = 110110$

$y = 000101$

$x \oplus y$

|   |   |   |
|---|---|---|
| 1 | 1 | 0 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 0 | 0 | 0 |

$|x \oplus y| = 4$

$\therefore \delta(x, y) = 4$

Example: Find the Hamming distance between  $x$  and  $y$ ;

$$(a) x = 110110 \quad ; y = 000101$$

$$(b) x = 001100 \quad ; y = 010110$$

$$x = 001100$$

$$y = 010110$$

$$x \oplus y = 011010$$

$$\delta(x, y) = |x \oplus y| = 3$$

Theorem: let

$x, y$  and  $z$  be the elements of  $B^m$ , then

(a)  $\delta\{x, y\} = \delta\{y, x\}$

(b)  $\delta\{x, y\} \geq 0$

(c)  $\delta\{x, y\} = 0$  if and only if  $x = y$  (case-1)

(d)  $\delta\{x, y\} \leq \delta\{x, \underline{z}\} + \delta\{\underline{z}, y\}$

$$\begin{aligned} \text{d) } \delta\{x, y\} &= |(x \oplus y)| = |x \oplus z \oplus z \oplus y| \leftarrow \\ &\leq |x \oplus z| + |y \oplus z| \leftarrow \\ &\leq \delta\{x, z\} + \delta\{z, y\} \end{aligned}$$



$$(i) \quad \delta(x, y) = \delta(y, x)$$

$$\begin{aligned} \delta(x, y) &= |x \oplus y| \\ &= |y \oplus x| \\ &= \delta(y, x). \end{aligned}$$

$$(ii) \quad \delta(x, y) \geq 0$$

case-1 If  $x = y$

$$\delta(x, x) = |x \oplus x| = 0$$

Case-11

If  $x, y$  ~~are~~ are  
different, then they  
are different for at least  
one position.

$$\Rightarrow \delta(x, y) = |x \oplus y|$$

~~⇒~~

⇒ The hamming distance  
is at least one.

$$\Rightarrow \delta(x, y) \geq 1$$

Case - 1, (1)  $\Rightarrow$

$$\delta(x, y) \geq 0.$$

Theorem: let

$x, y$  and  $z$  be the elements of  $B^m$ , then

(a)  $\delta\{x, y\} = \delta\{y, x\}$

(b)  $\delta\{x, y\} \geq 0$

(c)  $\delta\{x, y\} = 0$  if and only if  $x = y$

(d)  $\delta\{x, y\} \leq \delta\{x, z\} + \delta\{z, y\}$

$$x = 101001, y = 000110$$

$$\delta(x, y) = \delta(y, x); \quad \begin{array}{r} 101001x \\ 000110y \end{array}$$

$$\delta(x, y) = 1x \oplus y = x \oplus y$$

The minimum distance: The minimum distance of an encoding function  $e: B^m \rightarrow B^n$  is the minimum of the distances between all distinct pairs of code words; that is

$$\min\{\delta(e(x), e(y)) : x, y \in B^m\}$$

Example: consider the following (2,5) encoding function  $e$ :

$e(00) = 00000$

$e(10) = 10111$

$e(01) = 01110$

$e(11) = 11111$

$$\begin{array}{r} \oplus \\ 00000 \\ 10111 \\ \hline 10111 \\ \delta(2,7) = 4 \end{array}$$

$$\begin{array}{r} \oplus \\ 00000 \\ 01110 \\ \hline 01110 \\ \delta(2,7) = 3 \end{array}$$

Example: consider the following (2,5) encoding function  $e$ :

$$e(00) = 00000$$

$$e(10) = 10111$$

$$e(01) = 01110$$

$$e(11) = 11111$$

$$\begin{array}{r} 00000 \\ 10111 \\ \hline 10111 \end{array}$$

$$\begin{array}{r} 00000 \\ 01110 \\ \hline 01110 \\ \text{5(2,4)} \\ \equiv 3 \end{array}$$

$$\begin{array}{r} 00000 \\ 11111 \\ \hline 11111 \\ \text{5(2,4)} \\ \equiv 5 \end{array}$$

$$\begin{array}{r} 10111 \\ 01110 \\ \hline 10001 \end{array}$$

$$\begin{array}{r} \text{5(2,4)} \\ 11111 \\ \hline 00010 \\ \equiv 2 \end{array}$$

$$\begin{array}{r} 01110 \\ 11111 \\ \hline 10001 \\ \text{5(2,4)} \\ \equiv 5 \end{array}$$

Note: An  $(m, n)$  encoding function  $e: B^m \rightarrow B^n$  can detect  $k$  or fewer errors if and only if its minimum distance is at least  $k+1$

Consider the  $(3, 8)$  encoding function  $e: B^3 \rightarrow B^8$  defined by

$e(000) = 00000000$

$e(001) = 10111000$

$e(010) = 00101101$

$e(011) = 10010101$

$e(100) = 10100100$

$e(101) = 10001001$

$e(110) = 00011100$

$e(111) = 00110001$

How many errors will  $e$  detect.

Note: An  $(m, n)$  encoding function  $e: B^m \rightarrow B^n$  can detect k or fewer errors if and only if its minimum distance is at least  $k+1$

Solution: the minimum distance is 3 so  $e$  detect two or fewer errors

Group Code: An  $(m, n)$  encoding function

$e: B^m \rightarrow B^n$  is called a group code if

$N = e(B^m) = \{e(b): b \in B^m\} = \text{Range}(e)$  is a subgroup of  $B^n$

Example: consider the  $(3, 6)$  encoding function  $e: B^3 \rightarrow B^6$

$e(000) = \underline{000000}$

$e(001) = \underline{001100}$

$e(010) = \underline{010011}$

$e(011) = \underline{011111}$

$e(100) = \underline{100101}$

$e(101) = 101001$

$e(110) = 110110$

$e(111) = 111010$

Show that this encoding function is a group code:

Note: let  $e: B^m \rightarrow B^n$  is a group code. The minimum distance of  $e$  is the minimum weight of a non zero code word.



# Boolean matrix addition (mod-2 sum) and multiplication( mod-2 multiplication)

We have

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+1 & 0+1 & 1+0 & 1+1 \\ 0+1 & 1+1 & 1+0 & 0+1 \\ 1+0 & 0+1 & 0+1 & 1+1 \end{bmatrix} \\ = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}.$$

Observe that if  $\mathbf{F} = \mathbf{D} \oplus \mathbf{E}$ , then  $f_{ij}$  is zero when *both*  $d_{ij}$  and  $e_{ij}$  are zero or both are one. ■

# Boolean matrix addition (mod-2 sum) and multiplication (mod-2 multiplication)

| . | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |



We have

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



# Note:

Let  $\mathbf{D}$  and  $\mathbf{E}$  be  $m \times p$  Boolean matrices, and let  $\mathbf{F}$  be a  $p \times n$  Boolean matrix. Then

$$(\mathbf{D} \oplus \mathbf{E}) * \mathbf{F} = (\mathbf{D} * \mathbf{F}) \oplus (\mathbf{E} * \mathbf{F}).$$

That is, a distributive property holds for  $\oplus$  and  $*$ .



Let  $m < n$  and  $r = n - m$ . An  $n \times r$  Boolean matrix

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix},$$

$n - m = r$  rows

whose last  $r$  rows form the  $r \times r$  identity matrix, is called a **parity check matrix**. We use  $\mathbf{H}$  to define an encoding function  $e_H: B^m \rightarrow B^n$ . If  $b = b_1 b_2 \cdots b_m$ , let  $x = e_H(b) = \underline{b_1 b_2 \cdots b_m} \underline{x_1 x_2 \cdots x_r}$ , where

$$\begin{aligned} x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \cdots + b_m \cdot h_{m1} \\ x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \cdots + b_m \cdot h_{m2} \\ &\vdots \\ x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \cdots + b_m \cdot h_{mr}. \end{aligned} \tag{1}$$

**EXAMPLE 11**Let  $m = 2$ ,  $n = 5$ , and

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$e(\underline{B^2}) \rightarrow B^5$$

Determine the group code  $e_H: B^2 \rightarrow B^5$ .

$$e(11) = 11 \underline{x_1 x_2 x_3} = 11101$$

$$e(10) = 10 \underline{x_1 x_2 x_3} = 10110$$

$$e(01) = 01 \underline{x_1 x_2 x_3} = \checkmark$$

$$e(00) = 00 \underline{x_1 x_2 x_3} \checkmark$$

**Solution** We have  $B^2 = \{00, 10, 01, 11\}$ . Then

$$e(00) = 00x_1x_2x_3,$$

where  $x_1$ ,  $x_2$ , and  $x_3$  are determined by the equations in (1). Thus

$$x_1 = x_2 = x_3 = 0$$

and

$$e(00) = 00000.$$

Next

$$e(10) = 10x_1x_2x_3.$$

Using the equations in (1) with  $b_1 = 1$  and  $b_2 = 0$ , we obtain

$$x_1 = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$x_2 = 1 \cdot 1 + 0 \cdot 1 = 1$$

$$x_3 = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Thus  $x_1 = 1$ ,  $x_2 = 1$ , and  $x_3 = 0$ , so

$$e(10) = 10110. \quad \underline{\hspace{1cm}}$$

Similarly (verify),

$$e(01) = 01011 \quad \underline{\hspace{1cm}}$$

$$e(11) = 11101. \quad \underline{\hspace{1cm}}$$

# Decoding and error correction

Let  $e: B^m \rightarrow B^n$  is an encoding function that is encoded word  $x = e(b) \in B^n$ , for  $b \in B^m$  is received as the word  $x_t$ , by doing decoding we will receive  $x$ , for this we choose an onto function  $d: B^n \rightarrow B^m$  s.t  $d(x_t) = b' \in B^m$  is such that when the transmission channel has no noise, then  $b' = b$  that is  $d \circ e = I_{B^m}$  where  $I_{B^m}$  is the identity function of  $B^m$

Q.1 Find the minimum distance of the (2,4) encoding function  $e$  and how many errors will  $e$  detect?

$$e(0,0) = 0000$$

$$e(1,0) = 0110$$

$$e(0,1) = 1011$$

$$e(11) = 1100$$

identity element  $\swarrow$

| $\oplus$ | 0000 ✓ | 0110 ✓ | 1011 ✓      | 1100 ✓      |
|----------|--------|--------|-------------|-------------|
| 0000 ✓   | 0000   | 0110   | 1011        | 1100        |
| 0110     | 0110 ✓ | 0000 ✓ | <u>1101</u> | <u>1010</u> |
| 1011     | ✓      | ✓      | ✓           | ✓           |
| 1100     | ✓      | ✓      | ✓           | ✓           |

$$\begin{array}{r}
 1011 \\
 0110 \\
 \hline
 1101
 \end{array}$$

$$\begin{array}{r}
 1100 \\
 0110 \\
 \hline
 1010
 \end{array}$$

Show that the (2,5) encoding function

$e: B^2 \rightarrow B^5$  defined by

$$e(0,0) = 00000$$

$$e(1,0) = 10101$$

$$e(0,1) = 01110$$

$$e(11) = 11011$$

is a group code.

| $\oplus$ | 00000        | 10101        | 01110        | 11011        |
|----------|--------------|--------------|--------------|--------------|
| 00000    | <u>00000</u> | <u>10101</u> | <u>01110</u> | <u>11011</u> |
| 10101    | <u>10101</u> | <u>00000</u> | <u>11011</u> | <u>01110</u> |
| 01110    | <u>01110</u> | <u>11011</u> | <u>00000</u> | <u>10101</u> |
| 11011    | <u>11011</u> | <u>01110</u> | <u>10101</u> | <u>00000</u> |

$$\begin{array}{r}
 10101 \\
 11011 \\
 \hline
 01110
 \end{array}$$

$$\begin{array}{r}
 11011 \\
 01110 \\
 \hline
 10101
 \end{array}$$

Let  $H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  be a parity check

matrix. Determine the  $(2,5)$  group  
code  $e_H: B^2 \rightarrow B^5$